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Acoustic scattering by parallel plates with periodic connectors

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The scattering of a plane sound wave by two infinite, elastic, parallel plates constrained by equally spaced, identical, mechanical line connectors, which can support only compressional motion and which are simply hinged to both plates, is investigated using Fourier transforms. A formally exact solution is obtained and investigated asymptotically when the only fluid present occupies the half-space from which the sound wave is incident and provides heavy fluid loading to the plate, and likewise when this fluid also occupies the space between the plates. Conditions for resonant behaviour are also examined in these limits.

1. INTRODUCTION

Double-plate systems occur in many situations. They may be found, for example, in buildings, aircraft and marine structures. The frames used to connect and stiffen such double-wall constructions can be important sound bridges (Cremer *et al.* 1978), as can mechanical connections resulting from imperfect construction.

The problem of sound scattering by a single flexible surface with stiffening has received much attention. The simplifying assumption of infinite spatial periodicity facilitates the solution. Leppington (1978) provided a formally exact solution for a membrane or plate that is rigidly constrained along one or more parallel lines and evaluated it asymptotically in the limits of light or heavy fluid loading. A plate stiffened periodically by Euler beams was investigated by Mace (1980*a, b*) and evaluated numerically for heavy fluid loading and approximately for light fluid loading.

Infinite periodicity of connectors has also allowed the solution of the double-plate problem. Lin & Garrelick (1977) and Takahashi (1983) have provided formulations for calculating the sound transmitted through and scattered by infinite periodically connected plates. In each case their solutions have been evaluated only numerically, and for the case when the fluid surrounding and between the plates is air, which provides light fluid loading only, and when the structural connections are most significant.

The present work forms a sequel to the previous paper (Skelton 1990, hereafter referred to as I) in which the scattering of a plane wave by parallel elastic plates connected by a single mechanical connector was investigated in the asymptotic limit of heavy fluid loading. The parallel plates are here connected by spatially periodic connectors, which transmit only line forces to the plates (Evseev 1973),

but which can support longitudinal motion. The formally exact solution obtained is evaluated in the asymptotic limit, appropriate in some underwater applications, of heavy fluid loading on the plate and of both negligible fluid coupling and heavy fluid coupling between the plates.

2. THE MODEL PROBLEM

The system of parallel elastic plates and compressible fluids under consideration in this paper is the same as that described in I, but here the plates are connected by an infinite number of equally spaced identical connectors. Where possible the notation used in I is also used here. Thus plates 1 and 2 occupy the planes $z = 0$ and $z = -h$ respectively. The plates have bending stiffnesses $D_j = E_j h_j^3 / 12(1 - \sigma_j^2)$ and mass per unit area M_j , where h_j is the thickness of plate j , E_j is Young's modulus and σ_j is Poisson's ratio of the material of plate j ($j = 1, 2$). Compressible fluids occupy each of the regions separated by the plates: the half-space $z > 0$ is occupied by fluid 1, the region between the plates $-h < z < 0$ by fluid 2, and the half-space $z < -h$ is filled by fluid 3. The mean density of fluid j is ρ_j and its sound speed is c_j ($j = 1, 3$). An infinite number of connectors, which are each perpendicular to the plates and which can support compressional (z -direction) motion, are simply hinged to both plates along the lines $x = nL$, $z = 0$ and $x = nL$, $z = -h$ ($n = 0, \pm 1, \pm 2, \dots$). The conventional assumption (Lin & Garrelick 1977; Takahashi 1983; Skelton 1990) that the connectors are invisible to the fluid in region 2 is again made here. Thus standing waves cannot be built-up between adjacent connectors.

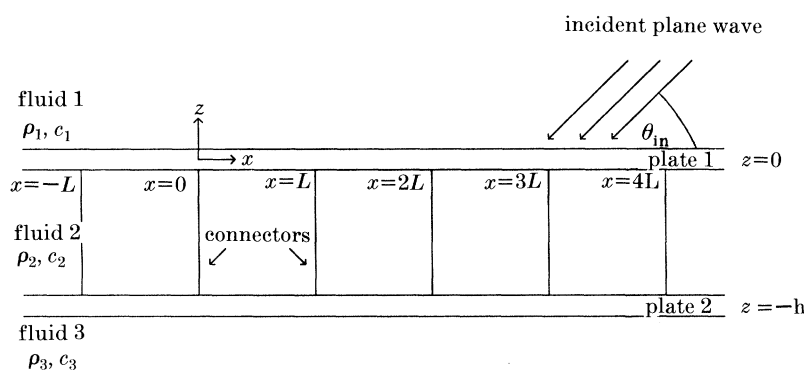


FIGURE 1. The scattering geometry.

A time-harmonic incident plane wave of frequency ω provides the system excitation,

$$p_{\text{in}}(x, z) = \exp[-ik_1(x \cos \theta_{\text{in}} + z \sin \theta_{\text{in}}) - i\omega t], \quad z > 0, \quad (2.1)$$

where $k_1 = \omega/c_1$, and θ_{in} is the angle between the plate and the propagation direction of the wave.

The equations governing the pressure fluctuations in each of the fluid regions are the wave equations:

$$(\partial^2/\partial x^2) p_j(x, z) + (\partial^2/\partial z^2) p_j(x, z) + k_j^2 p_j(x, z) = 0, \quad (2.2)$$

in which, $k_j = \omega/c_j$ and the pressures in fluids 2 and 3 are p_2 and p_3 respectively and, as in all subsequent equations, the time-harmonic factor $\exp(-i\omega t)$ has been omitted. The total pressure fluctuation, p_{TOT} , in fluid 1 is

$$p_{\text{TOT}}(x, z) = p_{\text{in}}(x, z) + p_r(x, z) + p_1(x, z), \quad z > 0, \quad (2.3)$$

where p_r is the wave in fluid 1 that would be reflected if plate 1 were rigid, thus

$$p_r(x, z) = \exp[-ik_1(x \cos \theta_{\text{in}} - z \sin \theta_{\text{in}})], \quad z > 0. \quad (2.4)$$

The thin-plate equation (see, for example, Cremer *et al.* 1988, p. 286) governs the transverse displacements of the plates:

$$D_1(\partial^4/\partial x^4) u_1(x) - \omega^2 M_1 u_1(x) = p_2(x, 0) - p_{\text{TOT}}(x, 0) + F_1(x) \quad (2.5)$$

and

$$D_2(\partial^4/\partial x^4) u_2(x) - \omega^2 M_2 u_2(x) = p_3(x, -h) - p_2(x, -h) + F_2(x), \quad (2.6)$$

in which $u_1(x)$ and $u_2(x)$ are the displacements of plate 1 and plate 2 respectively and $F_1(x)$ and $F_2(x)$ are the total reaction forces exerted on the plates by the infinite number of connectors. Thus

$$F_1(x) = \sum_{n=-\infty}^{\infty} F_{n1} \delta(x - nL), \quad (2.7)$$

$$F_2(x) = \sum_{n=-\infty}^{\infty} F_{n2} \delta(x - nL). \quad (2.8)$$

F_{n1} and F_{n2} are the amplitudes of the reaction forces exerted by the connector at $x = nL$ on plate 1 and plate 2 respectively. It has been assumed here that each connector is simply hinged to both plates in a way that prevents the transmission of moments to the plates. The compressional displacement of each connector satisfies the equation for longitudinal motion in a plate (see, for example, Cremer *et al.* 1988, p. 86), from which the relation between the reaction forces due to each connector and the displacements of the plates at each connector location is obtained as

$$\begin{bmatrix} F_{n1} \\ F_{n2} \end{bmatrix} = \begin{bmatrix} P & -Q \\ -Q & P \end{bmatrix} \begin{bmatrix} u_1(nL) \\ u_2(nL) \end{bmatrix}, \quad (2.9)$$

in which

$$P = -[E_c A/h(1 - \sigma_c^2)] \psi \cot \psi, \quad (2.10)$$

$$Q = -[E_c A/h(1 - \sigma_c^2)] \psi \operatorname{cosec} \psi, \quad (2.11)$$

$$\psi = \omega h(\rho_c(1 - \sigma_c^2)/E_c)^{\frac{1}{2}} \quad (2.12)$$

and E_c is Young's modulus, σ_c is Poisson's ratio and ρ_c is the density of the connector material, and A is the thickness of the connecting plate.

The equations that complete the description of this model are those that relate

the pressures in the fluids to the plate displacements by the requirement of continuity of normal velocity at the plate surfaces, i.e.

$$\partial p_1(x, z)/\partial z|_{z=0} = \rho_1 \omega^2 u_1(x), \quad (2.13)$$

$$\partial p_2(x, z)/\partial z|_{z=0} = \rho_2 \omega^2 u_1(x), \quad (2.14)$$

$$\partial p_2(x, z)/\partial z|_{z=-h} = \rho_2 \omega^2 u_2(x) \quad (2.15)$$

and

$$\partial p_3(x, z)/\partial z|_{z=-h} = \rho_3 \omega^2 u_2(x). \quad (2.16)$$

By considering the nature of the incident wave and the periodic geometry it is clear that the situation at any position x and time t is identical to that at position $x + nL$ at the earlier time $t - nL \cos \theta_{\text{in}}/c_1$. Hence, for example, the displacement of plate 1 satisfies

$$u_1(x + nL) = u_1(x) \exp[-ik_1 nL \cos \theta_{\text{in}}] \quad (n = 0, \pm 1, \pm 2, \dots), \quad (2.17)$$

whereas the reaction force amplitudes can each be related to the amplitudes of the reaction forces, F_1 and F_2 , of the connector at $x = 0$,

$$F_{n1} = F_1 \exp[-ik_1 nL \cos \theta_{\text{in}}] \quad (n = 0, \pm 1, \pm 2, \dots) \quad (2.18)$$

and

$$F_{n2} = F_2 \exp[-ik_1 nL \cos \theta_{\text{in}}] \quad (n = 0, \pm 1, \pm 2, \dots). \quad (2.19)$$

This allows the force distributions on the plates due to the connectors to be expressed as

$$F_1(x) = F_1 \sum_{n=-\infty}^{\infty} \exp[-ik_1 nL \cos \theta_{\text{in}}] \delta(x - nL), \quad (2.20)$$

$$F_2(x) = F_2 \sum_{n=-\infty}^{\infty} \exp[-ik_1 nL \cos \theta_{\text{in}}] \delta(x - nL). \quad (2.21)$$

Fourier transforms with respect to x can be defined for this problem. For example, the transform of the displacement of plate 1 is $\bar{u}_1(\alpha)$, where

$$\bar{u}_1(\alpha) = \int_{-\infty}^{\infty} u_1(x) e^{-i\alpha x} dx. \quad (2.22)$$

A method of obtaining a matrix equation relating the plate displacement transforms to the reaction force amplitudes is presented in I for the case of a single connector at $x = x_0$. This method, which is not repeated here, may also be applied to the case of periodic connectors. Alternatively, the required equation may be obtained directly from the single connector case by adding the terms due to the reaction forces with $x_0 = nL$ and with the appropriate phases obtained from equations (2.20) and (2.21). Thus,

$$S(\alpha) \begin{bmatrix} \bar{u}_1(\alpha) \\ \bar{u}_2(\alpha) \end{bmatrix} = -4\pi \begin{bmatrix} \delta(\alpha + k_1 \cos \theta_{\text{in}}) \\ 0 \end{bmatrix} + \begin{bmatrix} F_1 \\ F_2 \end{bmatrix} \sum_{n=-\infty}^{\infty} \exp[-inL(\alpha + k_1 \cos \theta_{\text{in}})], \quad (2.23)$$

where

$$S_{11}(\alpha) = D_1 \alpha^4 - \omega^2 M_1 - i \rho_1 \omega^2 / \gamma_1 + \rho_2 \omega^2 \cos(\gamma_2 h) / (\gamma_2 \sin(\gamma_2 h)), \quad (2.24)$$

$$S_{12}(\alpha) = -\rho_2 \omega^2 / (\gamma_2 \sin(\gamma_2 h)), \quad (2.25)$$

$$S_{21}(\alpha) = S_{12}(\alpha), \quad (2.26)$$

$$S_{22}(\alpha) = D_2 \alpha^4 - \omega^2 M_2 - i \rho_3 \omega^2 / \gamma_3 + \rho_2 \omega^2 \cos(\gamma_2 h) / (\gamma_2 \sin(\gamma_2 h)), \quad (2.27)$$

and the functions γ_j are defined for real values of α as

$$\gamma_j = \gamma_j(\alpha) = \begin{cases} \sqrt{k_j^2 - \alpha^2} & \text{if } k_j^2 \geq \alpha^2, \\ i\sqrt{\alpha^2 - k_j^2} & \text{if } k_j^2 \leq \alpha^2, \end{cases} \quad (2.28)$$

and by analytic continuation for complex values of α , with the branch cuts from $\alpha = \pm k_j$ to $\pm \infty$ chosen to ensure that

$$\text{Im}(\gamma_j) \geq 0 \quad \text{for complex } \alpha. \quad (2.29)$$

The matrix $S(\alpha)$ is the dynamic stiffness matrix for the system of two parallel elastic plates and three compressible fluids obtained in I. The first term on the right-hand side of equation (2.23) represents the excitation of the system of coupled fluids and plates (without the connecting plates), which is due to the incident plane wave. The second term on the right-hand side represents the additional excitation of the coupled system of fluids and parallel plates due to the reaction forces of the connectors. Periodicity of the geometry allows these forces to be expressed in terms of the (as yet unknown) reaction force amplitudes of the connector at $x = 0$. The analysis necessary to determine these amplitudes is presented in the next section.

3. THE REACTION FORCE AMPLITUDES

Extensive use will be made in the analysis that follows of Poisson's summation theorem, which states that (see, for example, Jones 1966):

If $g(x)$ is continuous and of bounded variation on $(-\infty, \infty)$ and tends to 0 as $|x| \rightarrow \infty$ such that

$$\int_{-\infty}^{\infty} g(x) dx$$

converges, then

$$\sum_{m=-\infty}^{\infty} g(\lambda m) = \frac{1}{\lambda} \sum_{n=-\infty}^{\infty} G(2n\pi/\lambda), \quad (3.1)$$

where

$$G(s) = \int_{-\infty}^{\infty} g(x) e^{-isx} dx. \quad (3.2)$$

By setting

$$g(x) = \exp[-ix(\alpha + k_1 \cos \theta_{\text{in}})], \quad (3.3)$$

for which

$$G(s) = 2\pi \delta(s + \alpha + k_1 \cos \theta_{\text{in}}), \quad (3.4)$$

Poisson's theorem can be used to rewrite the summation occurring on the right-hand side of equation (2.23) as

$$\sum_{n=-\infty}^{\infty} \exp[-inL(\alpha + k_1 \cos \theta_{\text{in}})] = \frac{2\pi}{L} \sum_{n=-\infty}^{\infty} \delta\left(\alpha + k_1 \cos \theta_{\text{in}} - \frac{2n\pi}{L}\right). \quad (3.5)$$

Substitution of this expression into equation (2.23) and evaluation of the inverse Fourier transform of $(\bar{u}_1(\alpha), \bar{u}_2(\alpha))^T$ obtained from the resulting equation leads to a relation between the plate displacements at $x = 0$ and the reaction force amplitudes there:

$$\begin{bmatrix} u_1(0) \\ u_2(0) \end{bmatrix} = \frac{1}{L} \begin{bmatrix} S_2 & S_3 \\ S_3 & S_1 \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix} - \frac{2}{\Delta(-k_1 \cos \theta_{\text{in}})} \begin{bmatrix} S_{22}(-k_1 \cos \theta_{\text{in}}) \\ -S_{12}(-k_1 \cos \theta_{\text{in}}) \end{bmatrix}, \quad (3.6)$$

in which $\Delta(\alpha)$ is the determinant of the matrix $S(\alpha)$, thus

$$\Delta(\alpha) = S_{11}(\alpha)S_{22}(\alpha) - S_{12}(\alpha)S_{21}(\alpha), \quad (3.7)$$

and

$$S_1 = \sum_{n=-\infty}^{\infty} \frac{S_{11}(2n\pi/L - k_1 \cos \theta_{\text{in}})}{\Delta(2n\pi/L - k_1 \cos \theta_{\text{in}})}, \quad (3.8)$$

$$S_2 = \sum_{n=-\infty}^{\infty} \frac{S_{22}(2n\pi/L - k_1 \cos \theta_{\text{in}})}{\Delta(2n\pi/L - k_1 \cos \theta_{\text{in}})} \quad (3.9)$$

and

$$S_3 = \sum_{n=-\infty}^{\infty} -\frac{S_{12}(2n\pi/L - k_1 \cos \theta_{\text{in}})}{\Delta(2n\pi/L - k_1 \cos \theta_{\text{in}})}. \quad (3.10)$$

Thus the integrals C_1, C_2 and B of the single connector solution (see I) are replaced for periodic connectors by the corresponding infinite summations S_1, S_2 and S_3 . In contrast to the single connector integrals C_1, C_2 and B , which are independent of θ_{in} , the sums S_1, S_2 and S_3 depend not only on the plate and fluid parameters, but also on $k_1 \cos \theta_{\text{in}}$ and the spacing of the connectors, L .

The force amplitudes are obtained by combining equations (3.6) and (2.9) as

$$\begin{bmatrix} F_1 \\ F_2 \end{bmatrix} = \left\{ \frac{-2}{W\Delta(-k_1 \cos \theta_{\text{in}})} \right\} [M] \begin{bmatrix} S_{22}(-k_1 \cos \theta_{\text{in}}) \\ -S_{12}(-k_1 \cos \theta_{\text{in}}) \end{bmatrix}, \quad (3.11)$$

where

$$M_{11} = P - (P^2 - Q^2)S_1/L, \quad (3.12)$$

$$M_{12} = -Q + (P^2 - Q^2)S_3/L, \quad (3.13)$$

$$M_{21} = M_{12}, \quad (3.14)$$

$$M_{22} = P - (P^2 - Q^2)S_2/L \quad (3.15)$$

and

$$W = 1 - [P(S_1 + S_2) - 2QS_3]/L + (P^2 - Q^2)(S_1S_2 - S_3^2)/L^2. \quad (3.16)$$

Hence expressions for the force amplitudes in terms of the physical parameters of the system of elastic plates and fluids are in principle available and they may be used to determine the plate displacements and the acoustic pressure in the compressible fluids.

4. THE PRESSURE IN FLUID 1

The total pressure in fluid 1 is given by equation (2.3) as the sum of the incident plane wave, a plane wave reflected as if the upper plate were rigid, and $p_1(x, z)$, where

$$p_1(x, z) = \frac{\rho_1 \omega^2}{2\pi i} \int_{-\infty}^{\infty} \frac{\bar{u}_1(\alpha) e^{i\alpha x} e^{i\gamma_1(\alpha)z}}{\gamma_1(\alpha)} d\alpha, \quad (4.1)$$

in which $\gamma_1(\alpha) = \sqrt{(k_1^2 - \alpha^2)}$ with the branch cuts defined by equations (2.28) and (2.29). When $\bar{u}_1(\alpha)$ is obtained from equations (2.23) and (3.5) this integral can be expressed in terms of F_1 and F_2 , themselves obtained from equation (3.11), as

$$p_1(x, z) = i\rho_1 \omega^2 \int_{-\infty}^{\infty} \left\{ 2S_{22}(\alpha) \delta(\alpha + k_1 \cos \theta_{\text{in}}) - \frac{F_1 S_{22}(\alpha) - F_2 S_{12}(\alpha)}{L} \sum_{n=-\infty}^{\infty} \delta\left(\alpha + k_1 \cos \theta_{\text{in}} - \frac{2n\pi}{L}\right) \right\} \frac{e^{i\alpha x} e^{i\gamma_1(\alpha)z}}{\gamma_1(\alpha) \Delta(\alpha)} d\alpha, \quad (4.2)$$

which can be evaluated exactly as

$$p_1(x, z) = i\rho_1 \omega^2 \left\{ \frac{2S_{22}(-k_1 \cos \theta_{\text{in}}) \exp(-ik_1 x \cos \theta_{\text{in}}) \exp(ik_1 z \sin \theta_{\text{in}})}{\Delta(-k_1 \cos \theta_{\text{in}}) k_1 \sin \theta_{\text{in}}} - \frac{F_1}{L} \sum_{n=-\infty}^{\infty} \frac{S_{22}(\kappa_n) \exp(i\kappa_n x + i\gamma_1(\kappa_n)z)}{\Delta(\kappa_n) \gamma_1(\kappa_n)} + \frac{F_2}{L} \sum_{n=-\infty}^{\infty} \frac{S_{12}(\kappa_n) \exp(i\kappa_n x + i\gamma_1(\kappa_n)z)}{\Delta(\kappa_n) \gamma_1(\kappa_n)} \right\}, \quad (4.3)$$

where

$$\kappa_n = 2n\pi/L - k_1 \cos \theta_{\text{in}}, \quad (4.4)$$

in both of the infinite summations in equation (4.3) the function $\gamma_1(\kappa_n)$ is real only for the finite number of terms for which

$$(\cos \theta_{\text{in}} - 1) k_1 L / 2\pi < n < (\cos \theta_{\text{in}} + 1) k_1 L / 2\pi. \quad (4.5)$$

As $z \rightarrow \infty$ the terms for which $\gamma_1(\kappa_n)$ are complex (they are in fact imaginary) are exponentially small and may be neglected in the evaluation of the far-field pressure. To each integer value of n within the range (4.5) there corresponds a scattered plane wave

$$(-i\rho_1 \omega^2 (F_1 S_{22}(k_1 \cos \theta_n) - F_2 S_{12}(k_1 \cos \theta_n)) \exp(ik_1 x \cos \theta_n) \exp(ik_1 z \sin \theta_n)) \div k_1 L \sin \theta_n \Delta(k_1 \cos \theta_n), \quad (4.6)$$

which propagates in the direction θ_n where

$$k_1 \cos \theta_n = \kappa_n. \quad (4.7)$$

In addition to these plane wave terms, which are caused directly by the periodic array of connectors, there is also a plane wave

$$\frac{2i\rho_1 \omega^2 S_{22}(-k_1 \cos \theta_{\text{in}}) \exp(-ik_1 x \cos \theta_{\text{in}}) \exp(ik_1 z \sin \theta_{\text{in}})}{k_1 \sin \theta_{\text{in}} \Delta(-k_1 \cos \theta_{\text{in}})}, \quad (4.8)$$

propagating in the direction of specular reflection. This wave would be present in the absence of the connectors and is due to the elastic properties of the coupled

fluid-plate system. In the far-field the acoustic pressure consists of a finite number of plane waves propagating in distinct directions. The character of the scattered sound field obtained by Leppington (1978) for a single membrane or plate with rigid line constraints was also of this type. Thus, as in the single connector case described in I, the pressure in fluid 1 can be decomposed into components identified as (i) the incident plane wave, (ii) a plane wave scattered as if plate 1 were rigid, (iii) a plane wave scattered in the specular direction due to the elastic vibration of the double plate system, which would be present in the absence of the connectors, and (iv) terms that can be attributed to the presence of the connectors. Unlike the single connector case, where the term due to the connector represents a cylindrically spreading disturbance with a continuous angular dependence, for a periodic array of connectors this term represents a finite number of plane waves propagating in well-defined directions. This occurs as a result of the interference effects of a combination of an infinite number of cylindrical waves, each of which is due to scattering by an individual connector.

In practice, infinitely long plates with infinitely many periodic connectors are not realizable. However, for sufficiently large (but finite length) plates with a sufficiently large (but finite) number of equally spaced connectors it is reasonable to assume that the large number of reaction forces will result, with some allowance for spreading and end effects, in almost plane wave scattering in the directions described above. Because these waves will be scattered only from the finite length of the plate, at large distances from the plate each scattered wave will be observable only in the particular direction in which it is propagating.

5. LEADING ORDER ASYMPTOTIC EVALUATION OF THE INFINITE SUMS WITH HEAVY FLUID LOADING ON PLATE 1: $\alpha_1 \gg k_1$, $\alpha_1 \gg (M_1 D_2 / M_2 D_1)^{1/2} \alpha_0$

(a) *No fluid coupling*: $\rho_2 = 0$

When $\rho_2 = 0$ there is no fluid coupling between the plates and the expressions (3.7)–(3.10), for the summations simplify. Thus

$$S_3 = 0 \quad (5.1)$$

and

$$A(\alpha) = S_{11}(\alpha) S_{22}(\alpha), \quad (5.2)$$

and hence

$$S_1 = \sum_{n=-\infty}^{\infty} \frac{1}{S_{22}(\kappa_n)}. \quad (5.3)$$

The Poisson summation theorem can be used again to evaluate this, by choosing

$$G(s) = 1/S_{22}(s - k_1 \cos \theta_{\text{in}}), \quad (5.4)$$

and thus

$$g(x) = \frac{\exp(ik_1 x \cos \theta_{\text{in}})}{2\pi} \int_{-\infty}^{\infty} \frac{e^{isx}}{S_{22}(s)} ds, \quad (5.5)$$

which, in the absence of fluid loading on plate 2, $\rho_3 = 0$, can be evaluated exactly as the sum of residues at α_0 and $i\alpha_0$ when $x \geq 0$ and as the sum of residues at $-\alpha_0$ and $-i\alpha_0$ when $x \leq 0$, as

$$g(x) = i \exp(ik_1 x \cos \theta_{\text{in}}) [\exp(i\alpha_0 |x|) + i \exp(-\alpha_0 |x|)] / 4D_2 \alpha_0^3, \quad (5.6)$$

in which α_0 is the *in vacuo* wavenumber of plate 2,

$$\alpha_0 = (\omega^2 M_2 / D_2)^{\frac{1}{4}}. \quad (5.7)$$

Poisson's summation theorem allows the sum S_1 to be rewritten as

$$S_1 = \frac{iL}{4D_2 \alpha_0^3} \sum_{n=-\infty}^{\infty} \exp(ik_1 Ln \cos \theta_{\text{in}}) [\exp(i\alpha_0 L|n|) + i \exp(-\alpha_0 L|n|)]. \quad (5.8)$$

These geometric series converge provided that the (small) imaginary part of α_0 due to damping in plate 2 is not less than the (small) imaginary part of $k_1 \cos \theta_{\text{in}}$ due to damping in fluid 1, and can be evaluated as

$$\sum_{n=-\infty}^{\infty} \exp(ik_1 Ln \cos \theta_{\text{in}}) \exp(i\alpha_0 L|n|) = \frac{-i \sin(\alpha_0 L)}{\cos(\alpha_0 L) - \cos(k_1 L \cos \theta_{\text{in}})} \quad (5.9)$$

and

$$\sum_{n=-\infty}^{\infty} \exp(ik_1 Ln \cos \theta_{\text{in}}) \exp(-\alpha_0 L|n|) = \frac{\sinh(\alpha_0 L)}{\cosh(\alpha_0 L) - \cos(k_1 L \cos \theta_{\text{in}})}. \quad (5.10)$$

Thus with negligible fluid loading on plate 2 the infinite sum S_1 is

$$S_1 = \frac{-L}{4D_2 \alpha_0^3} \left[\frac{\sin(\alpha_0 L)}{\cos(k_1 L \cos \theta_{\text{in}}) - \cos(\alpha_0 L)} - \frac{\sinh(\alpha_0 L)}{\cos(k_1 L \cos \theta_{\text{in}}) - \cosh(\alpha_0 L)} \right]. \quad (5.11)$$

An asymptotic approximation to the sum S_2 in the limit of heavy fluid loading on plate 1 is obtained in the same way. With no fluid coupling equation (3.9) simplifies to

$$S_2 = \sum_{n=-\infty}^{\infty} \frac{1}{S_{11}(\kappa_n)}. \quad (5.12)$$

Poisson's summation theorem can be applied to this with

$$G(s) = 1/S_{11}(s - k_1 \cos \theta_{\text{in}}) \quad (5.13)$$

and

$$g(x) = \frac{\exp(ik_1 x \cos \theta_{\text{in}})}{2\pi} \int_{-\infty}^{\infty} \frac{e^{isx}}{S_{11}(s)} ds. \quad (5.14)$$

When $x = 0$ the asymptotic form of this integral for heavy fluid loading has been evaluated in I. Thus

$$g(0) = [i - \cot(\frac{2}{5}\pi) + O(k_1^2/\alpha_1^2) + O((M_1 D_2/M_2 D_1)^{\frac{1}{2}} \alpha_0^2/\alpha_1^2)]/5D_1 \alpha_1^3, \quad (5.15)$$

where α_1 is the wavenumber of plate 1 with fluid loading on one side only,

$$\alpha_1 = (\rho_1 \omega^2 / D_1)^{\frac{1}{4}}. \quad (5.16)$$

When $x \neq 0$ the heavy fluid loading asymptotic form of the integral is obtained by first writing it as

$$\int_{-\infty}^{\infty} \frac{e^{isx}}{S_{11}(s)} ds = \int_{-\infty}^0 \frac{e^{isx}}{D_1 s^4 + \rho_1 \omega^2/s} ds + \int_0^{\infty} \frac{e^{isx}}{D_1 s^4 - \rho_1 \omega^2/s} ds + [O(k_1^2/\alpha_1^2) + O((M_1 D_2/M_2 D_1)^{1/2} \alpha_0^2/\alpha_1^2)]/D_1 \alpha_1^3, \quad (5.17)$$

and deforming the integration contours onto either the positive imaginary axis, if $x > 0$, or the negative imaginary axis, if $x < 0$. The leading-order term for heavy fluid loading is obtained as the residue from the pole on the real axis at

$$s = \alpha_1 \operatorname{sgn}(x), \quad (5.18)$$

which is crossed in deforming the contour. Thus

$$g(x) = \left\{ i \exp(ik_1 x \cos \theta_{\text{in}}) \exp(i\alpha_1 |x|) + O\left(\frac{k_1^2}{\alpha_1^2}\right) + O\left[\left(\frac{M_1 D_2}{M_2 D_1}\right)^{1/2} \frac{\alpha_0^2}{\alpha_1^2}\right] \right\} / 5D_1 \alpha_1^3, \quad x \neq 0. \quad (5.19)$$

The sum S_2 can therefore be rewritten, using Poisson's summation theorem, as

$$S_2 = \frac{L}{5D_1 \alpha_1^3} \left\{ -\cot\left(\frac{2}{5}\pi\right) + i \sum_{n=-\infty}^{\infty} \exp(ik_1 nL \cos \theta_{\text{in}}) \exp(i\alpha_1 L|n|) + O\left(\frac{k_1^2}{\alpha_1^2}\right) + O\left[\left(\frac{M_1 D_2}{M_2 D_1}\right)^{1/2} \frac{\alpha_0^2}{\alpha_1^2}\right] \right\} \quad (5.20)$$

in which the remaining infinite summation is of the same type as equation (5.9) and can be evaluated to give

$$S_2 = \frac{-L}{5D_1 \alpha_1^3} \left\{ \cot\left(\frac{2}{5}\pi\right) + \frac{\sin(\alpha_1 L)}{\cos(k_1 L \cos \theta_{\text{in}}) - \cos(\alpha_1 L)} + O\left(\frac{k_1^2}{\alpha_1^2}\right) + O\left[\left(\frac{M_1 D_2}{M_2 D_1}\right)^{1/2} \frac{\alpha_0^2}{\alpha_1^2}\right] \right\} \quad (5.21)$$

(b) *Heavy fluid coupling*: $\rho_2 = \rho_1$, $c_2 = c_1$

The approximations made above are not valid when the fluid in the region between the plates has significant mass. In this section this fluid is assumed to be the same as that in region 1, thus $\rho_1 = \rho_2$ and $c_1 = c_2$, and with heavy fluid loading on both plates. The heavy fluid loading assumption means that the approximate doubly fluid-loaded wavenumber of plate 1 is greater than both the wavenumber in the fluid and the *in vacuo* wavenumber of this plate, i.e. $2^{1/2} \alpha_1 \gg k_1$ and $2^{1/2} \alpha_1 \gg (M_1 D_2/M_2 D_1)^{1/2} \alpha_0$, and also that the approximate fluid-loaded wavenumber of plate 2 is greater than both the wavenumber in the fluid and the *in vacuo* wavenumber of this plate, i.e. $\alpha_2 \gg k_1$ and $\alpha_2 \gg \alpha_0$, where α_2 is the approximate fluid-loaded wavenumber of plate 2,

$$\alpha_2 = (\rho_1 \omega^2/D_2)^{1/2} \quad (5.22)$$

Thus for the fluid to provide heavy fluid loading to both plates its density ρ_1 must be large compared with each of

$$\frac{D_1}{2D_2} M_2 k_1 \left(\frac{k_1}{\alpha_0} \right)^2, \quad \frac{1}{2} \left(\frac{M_1 D_2}{M_2 D_1} \right)^{\frac{1}{4}} M_1 \alpha_0, \quad M_2 k_1 \left(\frac{k_1}{\alpha_0} \right)^4 \quad \text{and} \quad M_2 \alpha_0.$$

In addition to the heavy fluid loading assumptions of the coupling fluid the cases of a large or small separation between the plates will be considered separately.

(i) *Large plate separation*; $\alpha_1 h \gg 1$, $\alpha_2 h \gg 1$

As in paper I, large separation means here that

$$\exp(2h(2^{\frac{1}{2}}\alpha_1)) \gg 1 \quad (5.23)$$

and

$$\exp(2h\alpha_2) \gg 1, \quad (5.24)$$

to ensure that the fluid-loaded plate wavenumbers are included in the region of validity of the following large $|\alpha|$ approximations,

$$\sin(\gamma_2 h) \approx \frac{1}{2}i e^{|\alpha|h} \quad (5.25)$$

and

$$\cos(\gamma_2 h) \approx \frac{1}{2}e^{|\alpha|h}. \quad (5.26)$$

With significant fluid coupling the full equation (3.8) must be used to calculate S_1 . In this case the Poisson summation theorem can again be used with

$$G(s) = S_{11}(s - k_1 \cos \theta_{\text{in}}) / \Delta(s - k_1 \cos \theta_{\text{in}}), \quad (5.27)$$

and thus

$$g(x) = \frac{\exp(ik_1 x \cos \theta_{\text{in}})}{2\pi} \int_{-\infty}^{\infty} \frac{S_{11}(s)}{\Delta(s)} e^{isx} ds. \quad (5.28)$$

This integral has been evaluated in paper I when $x = 0$ in the asymptotic limit considered here, thus

$$g(0) = [i - \cot(\frac{2}{5}\pi) + O(k_1^2/\alpha_2^2) + O(\alpha_0^2/\alpha_2^2)] / 5D_2 \alpha_2^3 \quad (5.29)$$

When $x \neq 0$ the leading order term is obtained, as in the previous section, by approximating the integrand appropriately on two semi-infinite ranges and then deforming the integration contours onto either the positive or negative imaginary axes. The leading order term is found to be

$$g(x) = [i \exp(ik_1 x \cos \theta_{\text{in}}) \exp(i\alpha_2 |x|) + O(k_1^2/\alpha_2^2) + O(\alpha_0^2/\alpha_2^2)] / 5D_2 \alpha_2^3, \quad x \neq 0, \quad (5.30)$$

which is the residue from the pole at

$$s = \alpha_2 \operatorname{sgn}(x), \quad (5.31)$$

which is crossed in deforming the contour. In this asymptotic limit plate 2 behaves to leading order as if it were loaded by a semi-infinite fluid half-space and plate 1 were absent. The summation S_1 is, using Poisson's theorem,

$$S_1 = \frac{L}{5D_2\alpha_2^3} \left[-\cot\left(\frac{2}{5}\pi\right) + i \sum_{n=-\infty}^{\infty} \exp(ik_1 nL \cos \theta_{\text{in}}) \exp(i\alpha_2 L|n|) + O\left(\frac{k_1^2}{\alpha_2^2}\right) + O\left(\frac{\alpha_0^2}{\alpha_2^2}\right) \right] \quad (5.32)$$

that is,

$$S_1 = \frac{-L}{5D_2\alpha_2^3} \left[\cot\left(\frac{2}{5}\pi\right) + \frac{\sin(\alpha_2 L)}{\cos(k_1 L \cos \theta_{\text{in}}) - \cos(\alpha_2 L)} + O\left(\frac{k_1^2}{\alpha_2^2}\right) + O\left(\frac{\alpha_0^2}{\alpha_2^2}\right) \right]. \quad (5.33)$$

To calculate S_2 the Poisson summation theorem can be used with

$$G(s) = S_{22}(s - k_1 \cos \theta_{\text{in}}) / \Delta(s - k_1 \cos \theta_{\text{in}}) \quad (5.34)$$

and

$$g(x) = \frac{\exp(ik_1 x \cos \theta_{\text{in}})}{2\pi} \int_{-\infty}^{\infty} \frac{S_{22}(s)}{\Delta(s)} e^{isx} ds. \quad (5.35)$$

This integral has been evaluated in this asymptotic limit in paper I when $x = 0$. Thus

$$g(0) = \frac{1}{5D_1(2^{\frac{1}{5}}\alpha_1)^3} \left\{ i - \cot\left(\frac{2}{5}\pi\right) + O\left(\frac{k_1^2}{\alpha_1^2}\right) + O\left[\left(\frac{M_1 D_2}{M_2 D_1}\right)^{\frac{1}{2}} \frac{\alpha_0^2}{\alpha_1^2}\right] \right\}. \quad (5.36)$$

Plate 1 behaves to leading order in this asymptotic limit as if it were loaded on each side by a semi-infinite fluid half-space, and plate 2 were absent. The leading-order terms when $x \neq 0$ are obtained, as in previous sections, as the residue from the pole at

$$s = 2^{\frac{1}{5}}\alpha_1 \operatorname{sgn}(x), \quad (5.37)$$

which is crossed when the integration contours are deformed. Thus

$$g(x) = \left\{ i \exp(ik_1 x \cos \theta_{\text{in}}) \exp(i2^{\frac{1}{5}}\alpha_1 |x|) + O(k_1^2/\alpha_1^2) + O\left[\left(\frac{M_1 D_2}{M_2 D_1}\right)^{\frac{1}{2}} \frac{\alpha_0^2}{\alpha_1^2}\right] \right\} / 5D_1(2^{\frac{1}{5}}\alpha_1)^3, \quad x \neq 0, \quad (5.38)$$

and the sum S_2 is obtained as

$$S_2 = \frac{-L}{5D_1(2^{\frac{1}{5}}\alpha_1)^3} \left\{ \cot\left(\frac{2}{5}\pi\right) + \frac{\sin(2^{\frac{1}{5}}\alpha_1 L)}{\cos(k_1 L \cos \theta_{\text{in}}) - \cos(2^{\frac{1}{5}}\alpha_1 L)} + O\left(\frac{k_1^2}{\alpha_1^2}\right) + O\left[\left(\frac{M_1 D_2}{M_2 D_1}\right)^{\frac{1}{2}} \frac{\alpha_0^2}{\alpha_1^2}\right] \right\} \quad (5.39)$$

To obtain S_3 it is necessary to set

$$G(s) = S_{12}(s - k_1 \cos \theta_{\text{in}}) / \Delta(s - k_1 \cos \theta_{\text{in}}). \quad (5.40)$$

As noted in I,

$$g(0) = [O(k_1^2/\alpha_2^2) + O(\alpha_0^2/\alpha_2^2)]/D_2\alpha_2^3, \quad (5.41)$$

which is of the same order as terms already neglected in the summations S_1 and S_2 . Similarly when $x \neq 0$

$$g(x) = [O(k_1^2/\alpha_2^2) + O(\alpha_0^2/\alpha_2^2)]/D_2 \alpha_2^3, \quad (5.42)$$

and thus

$$S_3 = [O(k_1^2/\alpha_2^2) + O(\alpha_0^2/\alpha_2^2)]/D_2 \alpha_2^3, \quad (5.43)$$

which may be neglected.

(ii) *Small plate separation*: $\alpha_1 h \ll 1$, $\alpha_2 h \ll 1$

Small plate separation means here, as in I, that the fluid-loaded plate wavenumbers both lie in the region where the trigonometric functions can be approximated by their small argument expansions. That is,

$$\sin(\gamma_2 h) \approx i\alpha h, \quad (5.44)$$

$$\cos(\gamma_2 h) \approx 1, \quad (5.45)$$

i.e.

$$\alpha_1 h \ll 1, \quad (5.46)$$

$$\alpha_2 h \ll 1. \quad (5.47)$$

Each of the sums S_1 , S_2 and S_3 can also be evaluated in this asymptotic limit by using the Poisson summation theorem. Thus, for example, the function $G(s)$ defined by equation (5.40) can be used to evaluate S_3 . When $x = 0$ the leading-order terms of its inverse, $g(0)$, have been obtained in I as the sum of residues from the pole on the real axis located at

$$s \approx \bar{\alpha} = \alpha_{12} [1 + \frac{1}{5}(1 - (\alpha_{12}/\alpha_2)^{10}) \alpha_{12} h], \quad (5.48)$$

where α_{12} is the wavenumber of a heavily fluid-loaded plate whose bending stiffness is $D_1 + D_2$,

$$\alpha_{12} = (\rho_1 \omega^2 / (D_1 + D_2))^{\frac{1}{5}} \quad (5.49)$$

and from the pole at

$$s \approx \tilde{\alpha} = [\alpha_1^5 \alpha_2^5 / \alpha_{12}^5 h]^{\frac{1}{6}} \quad (5.50)$$

It was shown in I that

$$g(0) = \left\{ \frac{1}{5} [i - \cot(\frac{2}{5}\pi)] - \frac{1}{6} i \left(\frac{\alpha_{12}}{(\alpha_1 \alpha_2)^{\frac{1}{2}}} \right)^5 (\alpha_{12} h)^{\frac{1}{2}} + O(\alpha_{12} h) \right. \\ \left. + O\left[\left(\frac{k_1}{\alpha_{12}} \right)^2 \right] + O\left[\left(\frac{1 + M_1/M_2}{1 + D_1/D_2} \right)^{\frac{1}{2}} \left(\frac{\alpha_0}{\alpha_{12}} \right)^2 \right] \right\} / (D_1 + D_2) \alpha_{12}^3. \quad (5.51)$$

when $x \neq 0$ the leading-order terms in the asymptotic expansion in this limit are obtained as the sum of the residues from the poles at

$$s = \bar{\alpha} \operatorname{sgn}(x) \quad (5.52)$$

and

$$s = \tilde{\alpha} \operatorname{sgn}(x), \quad (5.53)$$

which are crossed when the integration contours are deformed as described earlier. Thus one finds that

$$g(x) = \frac{i \exp(i k_1 x \cos \theta_{\text{in}})}{(D_1 + D_2) \alpha_{12}^3} \left\{ \frac{1}{5} \exp(i \tilde{\alpha} |x|) - \left[\frac{\alpha_{12}}{(\alpha_1 \alpha_2)^{\frac{1}{2}}} \right]^5 (\alpha_{12} h)^{\frac{1}{2}} \exp(i \tilde{\alpha} |x|) + O(\alpha_{12} h) \right. \\ \left. + O\left[\left(\frac{k_1}{\alpha_{12}}\right)^2\right] + O\left[\left(\frac{1+M_1/M_2}{1+D_1/D_2}\right)^{\frac{1}{2}} \left(\frac{\alpha_0}{\alpha_{12}}\right)^2\right] \right\}, \quad x \neq 0. \quad (5.54)$$

Both the $O(1)$ and $O((\alpha_{12} h)^{\frac{1}{2}})$ terms above are of the type obtained earlier and the sum S_3 may be evaluated directly as

$$S_3 = \frac{-L}{(D_1 + D_2) \alpha_{12}^3} \left\{ \frac{1}{5} \cot\left(\frac{2}{5}\pi\right) + \frac{\sin(\tilde{\alpha} L)}{5[\cos(k_1 L \cos \theta_{\text{in}}) - \cos(\tilde{\alpha} L)]} \right. \\ \left. - \left[\frac{\alpha_{12}}{(\alpha_1 \alpha_2)^{\frac{1}{2}}} \right]^5 \frac{(\alpha_{12} h)^{\frac{1}{2}} \sin(\tilde{\alpha} L)}{6[\cos(k_1 L \cos \theta_{\text{in}}) - \cos(\tilde{\alpha} L)]} + O(\alpha_{12} h) + O\left[\left(\frac{k_1}{\alpha_{12}}\right)^2\right] \right. \\ \left. + O\left[\left(\frac{1+M_1/M_2}{1+D_1/D_2}\right)^{\frac{1}{2}} \left(\frac{\alpha_0}{\alpha_{12}}\right)^2\right] \right\}. \quad (5.55)$$

The sums S_1 and S_2 are obtained similarly as

$$S_1 = \frac{-L}{(D_1 + D_2) \alpha_{12}^3} \left\{ \frac{1}{5} \cot\left(\frac{2}{5}\pi\right) + \frac{\sin(\tilde{\alpha} L)}{5[\cos(k_1 L \cos \theta_{\text{in}}) - \cos(\tilde{\alpha} L)]} \right. \\ \left. + \left(\frac{\alpha_2}{\alpha_1}\right)^5 \left[\frac{\alpha_{12}}{(\alpha_1 \alpha_2)^{\frac{1}{2}}} \right]^5 \frac{(\alpha_{12} h)^{\frac{1}{2}} \sin(\tilde{\alpha} L)}{6[\cos(k_1 L \cos \theta_{\text{in}}) - \cos(\tilde{\alpha} L)]} + O(\alpha_{12} h) \right. \\ \left. + O\left[\left(\frac{k_1}{\alpha_{12}}\right)^2\right] + O\left[\left(\frac{1+M_1/M_2}{1+D_1/D_2}\right)^{\frac{1}{2}} \left(\frac{\alpha_0}{\alpha_{12}}\right)^2\right] \right\} \quad (5.56)$$

and

$$S_2 = \frac{-L}{(D_1 + D_2) \alpha_{12}^3} \left\{ \frac{1}{5} \cot\left(\frac{2}{5}\pi\right) + \frac{\sin(\tilde{\alpha} L)}{5[\cos(k_1 L \cos \theta_{\text{in}}) - \cos(\tilde{\alpha} L)]} \right. \\ \left. + \left(\frac{\alpha_1}{\alpha_2}\right)^5 \left[\frac{\alpha_{12}}{(\alpha_1 \alpha_2)^{\frac{1}{2}}} \right]^5 \frac{(\alpha_{12} h)^{\frac{1}{2}} \sin(\tilde{\alpha} L)}{6[\cos(k_1 L \cos \theta_{\text{in}}) - \cos(\tilde{\alpha} L)]} + O(\alpha_{12} h) \right. \\ \left. + O\left[\left(\frac{k_1}{\alpha_{12}}\right)^2\right] + O\left[\left(\frac{1+M_1/M_2}{1+D_1/D_2}\right)^{\frac{1}{2}} \left(\frac{\alpha_0}{\alpha_{12}}\right)^2\right] \right\}. \quad (5.57)$$

Thus with a small plate separation and heavy fluid coupling the leading-order behaviour of the two plates is that of a single plate with bending stiffness $(D_1 + D_2)$ with fluid loading on one side only and with periodic constraints. The $O((\alpha_{12} h)^{\frac{1}{2}})$ correction terms represent the differences from the single plate behaviour due to the compressibility of the fluid layer.

6. CONDITIONS FOR RESONANCE

Resonant behaviour of the system, as reflected in the behaviour of either the far-field scattered pressures, given by equation (4.6), or the plate displacement, can occur only if

$$F_1 S_{22}(k_1 \cos \theta_n) - F_2 S_{12}(k_1 \cos \theta_n) \rightarrow \infty \quad (6.1)$$

for values of θ_n defined by equations (4.4) and (4.7), where the force amplitudes are those determined from the matrix equation (3.11). A necessary condition for resonance is that the force amplitudes should become infinite. Equations (3.11) and (3.16) indicate that this occurs if

$$1 - [P(S_1 + S_2) - 2QS_3]/L + (P^2 - Q^2)(S_1 S_2 - S_3^2)/L^2 = 0. \quad (6.2)$$

In the analysis that follows, attention is restricted to connectors that are both light (massless) and rigid. The resonance condition in this case is

$$S_1 + S_2 - 2S_3 = 0, \quad (6.3)$$

which may be obtained from equation (3.6) by setting

$$F_1 + F_2 = 0 \quad (6.4)$$

and

$$u_1(0) = u_2(0), \quad (6.5)$$

or from the general resonance requirement (6.2) in the limiting case when

$$P \sim Q \rightarrow \infty. \quad (6.6)$$

(a) *No fluid coupling*: $\rho_2 = 0$

In the absence of fluid coupling $\rho_2 = 0$ and equation (6.3) simplifies further to

$$S_1 + S_2 = 0. \quad (6.7)$$

It is convenient to define ϕ , $0 \leq \phi \leq \pi$, such that

$$\cos \phi = \cos(k_1 L \cos \theta_{in}), \quad (6.8)$$

and to note the following identities

$$\frac{\sin(\alpha_0 L)}{\cos(k_1 L \cos \theta_{in}) - \cos(\alpha_0 L)} = \frac{1}{2}[\cot(\frac{1}{2}(\alpha_0 L - \phi)) + \cot(\frac{1}{2}(\alpha_0 L + \phi))], \quad (6.9)$$

$$\frac{\sinh(\alpha_0 L)}{\cos(k_1 L \cos \theta_{in}) - \cosh(\alpha_0 L)} = -\frac{1}{2}[\coth(\frac{1}{2}(\alpha_0 L - i\phi)) + \coth(\frac{1}{2}(\alpha_0 L + i\phi))]. \quad (6.10)$$

When these are applied to the expressions (5.11) and (5.21) for S_1 and S_2 respectively the leading-order term of the resonance requirement (6.7) can be written as

$$\begin{aligned} & \frac{L}{4D_2 \alpha_0^3} [\cot(\frac{1}{2}(\alpha_0 L - \phi)) + \cot(\frac{1}{2}(\alpha_0 L + \phi)) + \coth(\frac{1}{2}(\alpha_0 L - i\phi)) + \coth(\frac{1}{2}(\alpha_0 L + i\phi))] \\ &= -\frac{L}{5D_1 \alpha_1^3} [\cot(\frac{1}{2}(\alpha_1 L - \phi)) + \cot(\frac{1}{2}(\alpha_1 L + \phi)) + 2 \cot(\frac{2}{5}\pi)]. \end{aligned} \quad (6.11)$$

As a function of α_0 , with ϕ fixed, the left-hand side of equation (6.11) is continuous in each of the intervals

$$2m\pi - \phi < \alpha_0 L < 2m\pi + \phi, \quad m = 1, 2, 3, \dots, \quad (6.12)$$

$$2m\pi + \phi < \alpha_0 L < 2(m+1)\pi - \phi, \quad m = 0, 1, 2, 3, \dots, \quad (6.13)$$

and varies from $-\infty$ at the lower end to $+\infty$ at the upper end of each interval. Similarly, as a function of α_1 , with ϕ fixed, the right-hand side of equation (6.11) is continuous in each of the intervals

$$2m\pi - \phi < \alpha_1 L < 2m\pi + \phi, \quad m = 1, 2, 3, \dots, \quad (6.14)$$

$$2m\pi + \phi < \alpha_1 L < 2(m+1)\pi - \phi, \quad m = 0, 1, 2, 3, \dots, \quad (6.15)$$

and varies from $+\infty$ at the lower end of each interval to $-\infty$ at the upper end. In terms of the frequency ω the intervals (6.12) and (6.13) can be expressed as

$$\left[\frac{2m\pi - \phi}{L} \right]^2 \left(\frac{D_2}{M_2} \right)^{\frac{1}{2}} < \omega < \left[\frac{2m\pi + \phi}{L} \right]^2 \left(\frac{D_2}{M_2} \right)^{\frac{1}{2}}, \quad m = 1, 2, 3, \dots, \quad (6.16)$$

and

$$\left[\frac{2m\pi + \phi}{L} \right]^2 \left(\frac{D_2}{M_2} \right)^{\frac{1}{2}} < \omega < \left[\frac{2(m+1)\pi - \phi}{L} \right]^2 \left(\frac{D_2}{M_2} \right)^{\frac{1}{2}}, \quad m = 0, 1, 2, 3, \dots, \quad (6.17)$$

and those of equations (6.14) and (6.15) as

$$\left[\frac{2m\pi - \phi}{L} \right]^{\frac{5}{2}} \left(\frac{D_1}{\rho_1} \right)^{\frac{1}{2}} < \omega < \left[\frac{2m\pi + \phi}{L} \right]^{\frac{5}{2}} \left(\frac{D_1}{\rho_1} \right)^{\frac{1}{2}}, \quad m = 1, 2, 3, \dots, \quad (6.18)$$

and

$$\left[\frac{2m\pi + \phi}{L} \right]^{\frac{5}{2}} \left(\frac{D_1}{\rho_1} \right)^{\frac{1}{2}} < \omega < \left[\frac{2(m+1)\pi - \phi}{L} \right]^{\frac{5}{2}} \left(\frac{D_1}{\rho_1} \right)^{\frac{1}{2}}, \quad m = 0, 1, 2, 3, \dots \quad (6.19)$$

In each of the intervals (6.16) and (6.17) the left-hand side of equation (6.11) varies continuously from $-\infty$ to $+\infty$, whereas the right-hand side is continuous except at a finite number of discontinuities at which it tends to $-\infty$ when approached from below, and $+\infty$ when approached from above. This is sufficient to ensure that at least one solution of equation (6.11) occurs within each of the frequency ranges defined by inequalities (6.16) and (6.17). A similar argument can be applied to show also that at least one solution of equation (6.11) occurs within each of the frequency ranges defined by inequalities (6.18) and (6.19). Thus, for each value of ϕ there are infinitely many frequencies at which resonant behaviour can occur. At each of these frequencies there are a finite number of values of the incident angle, θ_{in} , that satisfy equation (6.8), which defines ϕ . These values of θ_{in} satisfy

$$\theta_{\text{in}} = \arccos[(2m\pi + \phi)/k_1 L] \quad (6.20)$$

for integer values of m such that

$$-\phi - k_1 L < 2m\pi < -\phi + k_1 L \quad (6.21)$$

or

$$\theta_{\text{in}} = \arccos[(2m\pi - \phi)/k_1 L] \quad (6.22)$$

for integer values of m satisfying

$$\phi - k_1 L < 2m\pi < \phi + k_1 L. \quad (6.23)$$

Thus for any value of ϕ resonant behaviour can occur at an infinite number of frequencies, but for each such frequency resonance occurs only when the incident angle is one of a finite number of values associated with that frequency. Hence, as ϕ varies the pairs of values of frequency and incident angle at which resonance can occur also vary.

The resonance condition (6.11) can be simplified in the limits of low or very high frequencies. At low frequencies where $\alpha_0 \ll \alpha_1$ it becomes

$$\frac{\sin(\alpha_0 L)}{\cos \phi - \cos(\alpha_0 L)} - \frac{\sinh(\alpha_0 L)}{\cos \phi - \cosh(\alpha_0 L)} = 0, \quad (6.24)$$

and if $\alpha_0 L$ is sufficiently large this can be approximated further as

$$\frac{\sin(\alpha_0 L)}{\cos \phi - \cos(\alpha_0 L)} + 1 = 0 \quad (6.25)$$

or

$$\cos(\alpha_0 L + \frac{1}{4}\pi) = \frac{1}{\sqrt{2}} \cos \phi. \quad (6.26)$$

Thus as ϕ varies in this limit the solutions are restricted to the ranges

$$m\pi < \alpha_0 L < m\pi + \frac{1}{2}\pi. \quad (6.27)$$

At very high frequencies, where $\alpha_1 \ll \alpha_0$, which may not in fact be of interest physically because the heavy fluid loading assumption that $\alpha_1 \gg (M_1 D_2 / M_2 D_1)^{\frac{1}{2}} \alpha_0$ imposes an upper limit on the frequency and can be simultaneously satisfied only if $(M_1 D_2 / M_2 D_1)^{\frac{1}{2}} \ll 1$, the behaviour of equation (6.11) can nevertheless be deduced by approximating it as

$$\frac{\sin(\alpha_1 L)}{\cos \phi - \cos(\alpha_1 L)} + \cos(\frac{2}{5}\pi) = 0 \quad (6.28)$$

or

$$\cos(\alpha_1 L + \frac{2}{5}\pi) = \cos(\frac{2}{5}\pi) \cos \phi, \quad (6.29)$$

for which solutions are confined to the ranges

$$m\pi < \alpha_1 L < m\pi + \frac{1}{5}\pi. \quad (6.30)$$

Thus at the extremes of both low and very high frequencies, frequency bands exist for which resonance, as defined by equation (6.11), cannot occur for any value of the incident angle.

(b) *Heavy fluid coupling*: $\rho_2 = \rho_1$, $c_2 = c_1$

(i) *Large plate separation*: $\alpha_1 h \gg 1$, $\alpha_2 h \gg 1$

When the heavy fluid 1 also occupies the region between the plates, $\rho_1 = \rho_2$ and $c_1 = c_2$, the resonance condition (6.3) can be written to leading order using the appropriate asymptotic expansions for S_1 , S_2 and S_3 given by equations (5.33), (5.39) and (5.43) respectively as

$$\begin{aligned} \frac{L}{5D_2\alpha_2^3} \left[\cot(\frac{2}{5}\pi) + \frac{\sin(\alpha_2 L)}{\cos(k_1 L \cos \theta_{\text{in}}) - \cos(\alpha_2 L)} \right] \\ = \frac{-L}{5D_1(2^{\frac{1}{5}}\alpha_1)^3} \left[\cot(\frac{2}{5}\pi) + \frac{\sin(2^{\frac{1}{5}}\alpha_1 L)}{\cos(k_1 L \cos \theta_{\text{in}}) - \cos(2^{\frac{1}{5}}\alpha_1 L)} \right] \end{aligned} \quad (6.31)$$

or

$$[2 \cot(\frac{2}{5}\pi) + \cot(\frac{1}{2}(\alpha_2 L - \phi)) + \cot(\frac{1}{2}(\alpha_2 L + \phi))] \\ = -\frac{1}{2} \left(\frac{2D_2}{D_1} \right)^{\frac{2}{5}} [2 \cot(\frac{2}{5}\pi) + \cot(\frac{1}{2}(2^{\frac{1}{5}}\alpha_1 L - \phi)) + \cot(\frac{1}{2}(2^{\frac{1}{5}}\alpha_1 L + \phi))]. \quad (6.32)$$

This equation has similar properties to those of equation (6.11) and the same arguments can be used to demonstrate that there is at least one solution in each of the frequency intervals

$$\left(\frac{2m\pi - \phi}{L} \right)^{\frac{5}{2}} \left(\frac{D_1}{2\rho_1} \right)^{\frac{1}{2}} < \omega < \left(\frac{2m\pi + \phi}{L} \right)^{\frac{5}{2}} \left(\frac{D_1}{2\rho_1} \right)^{\frac{1}{2}}, \quad m = 1, 2, 3, \dots, \quad (6.33)$$

and

$$\left(\frac{2m\pi + \phi}{L} \right)^{\frac{5}{2}} \left(\frac{D_1}{2\rho_1} \right)^{\frac{1}{2}} < \omega < \left(\frac{2(m+1)\pi - \phi}{L} \right)^{\frac{5}{2}} \left(\frac{D_1}{2\rho_1} \right)^{\frac{1}{2}}, \quad m = 0, 1, 2, 3, \dots, \quad (6.34)$$

and also in each of the intervals

$$\left(\frac{2m\pi - \phi}{L} \right)^{\frac{5}{2}} \left(\frac{D_2}{\rho_1} \right)^{\frac{1}{2}} < \omega < \left(\frac{2m\pi + \phi}{L} \right)^{\frac{5}{2}} \left(\frac{D_2}{\rho_1} \right)^{\frac{1}{2}}, \quad m = 1, 2, 3, \dots, \quad (6.35)$$

and

$$\left(\frac{2m\pi + \phi}{L} \right)^{\frac{5}{2}} \left(\frac{D_2}{\rho_1} \right)^{\frac{1}{2}} < \omega < \left(\frac{2(m+1)\pi - \phi}{L} \right)^{\frac{5}{2}} \left(\frac{D_2}{\rho_1} \right)^{\frac{1}{2}}, \quad m = 0, 1, 2, 3, \dots \quad (6.36)$$

Thus when the plates are well separated and the heavy fluid occupies the region between the plates there are infinitely many frequencies at which resonant behaviour may occur and for each frequency resonance can only occur at the finite number of angles of incidence given by equations (6.20)–(6.23), as in the case when there is no fluid coupling.

(ii) *Small plate separation*: $\alpha_1 h \ll 1$, $\alpha_2 h \ll 1$

When the plates are closely spaced and the heavy fluid occupies the region between them the necessary condition for resonance is obtained, to leading order, from equation (6.3), using the asymptotic expansions for the sums S_1 , S_2 and S_3 given in this limit by equations (5.56), (5.57) and (5.55) respectively, as

$$\frac{-L}{6(D_1 D_2)^{\frac{1}{2}} \alpha_{12}^3} \left[\frac{(\alpha_{12} h)^{\frac{1}{2}} \sin(\tilde{\alpha} L)}{[\cos(k_1 L \cos \theta_{\text{in}}) - \cos(\tilde{\alpha} L)]} \right] = 0, \quad (6.37)$$

which can be satisfied only if

$$\tilde{\alpha} L = n\pi, \quad (6.38)$$

or, in terms of frequency,

$$\omega = \left(\frac{h D_1 D_2}{\rho_1 (D_1 + D_2)} \right)^{\frac{1}{2}} \left(\frac{n\pi}{L} \right)^3, \quad (6.39)$$

which is independent of ϕ and thus in this limit the resonance can occur for all values of the incident angle at such frequencies.

7. AMPLITUDES AT RESONANCE: $\rho_2 = 0$

In the previous section resonance was said to occur when pairs of values for frequency and incident angle were such that the leading-order terms of the asymptotic expansions of certain infinite summations satisfied the resonance condition (6.3). In fact, at these values the force amplitudes, plate displacements and scattered pressure all remain finite because the requirement (6.3) has not been satisfied exactly, but only to leading order, and the higher-order terms in the expansions evaluated at these values therefore determine the amplitudes of the reaction forces, plate displacements and scattered pressure at the 'resonance'. In each of the asymptotic limits considered herein the leading-order terms that occur in (6.3) are real for real values of the frequency. The higher-order terms may, however, be complex. At the 'resonance' the real part of the higher-order terms can be interpreted as providing a small correction to the frequency at which resonance occurs. However, the imaginary part of the higher-order terms at the resonance remains non-zero and can be used to determine the amplitude of the resonance.

In the analysis that follows attention will be restricted to the situation in which there is no fluid between the plates, considered previously in §5(a) and §6(a). In this case, in which there is no fluid loading on plate 2, the evaluation of S_1 given in equation (5.11) is exact, and no further analysis of this term is required. The expression (5.21) given previously for S_2 is, however, the leading-order term only. To obtain the higher-order terms in the asymptotic expansion of S_2 , higher-order terms in the evaluation of $g(x)$ defined by equation (5.14) are required. In Appendix A these are obtained as

$$g(0) = [i - \cot(\frac{2}{5}\pi) + (k_1/\alpha_1)^2 \{5 \ln(2\alpha_1/k_1) - 1 + \frac{3}{2}i\pi\}/2\pi]/5D_1 \alpha_1^3 + \text{higher-order terms} \quad (7.1)$$

and, when $x \neq 0$ as,

$$\begin{aligned} g(x) = & i \exp(ik_1 x \cos \theta_{\text{in}}) [(1 - \frac{1}{2}(k_1/\alpha_1)^2) \exp(i\alpha_1(1 + k_1^2/10\alpha_1^2)|x|) \\ & + (5k_1/2\alpha_1^2|x|)(J_1(k_1|x|) + iY_1(k_1|x|))]/5D_1 \alpha_1^3 \\ & + \text{higher-order terms, } x \neq 0. \end{aligned} \quad (7.2)$$

The next-order terms in the asymptotic expansion of S_2 can be obtained by applying Poisson's summation theorem to these expressions to get

$$\begin{aligned} S_2 = & -L \left[\cot(\frac{2}{5}\pi) + \frac{(1 - \frac{1}{2}(k_1/\alpha_1)^2) \sin(\bar{\alpha}L)}{\cos(k_1 L \cos \theta_{\text{in}}) - \cos(\bar{\alpha}L)} - \left(\frac{k_1}{\alpha_1}\right)^2 \left\{ 5 \ln\left(\frac{2\alpha_1}{k_1}\right) - 1 + \frac{5i\pi}{2} \right\} / 2\pi \right. \\ & \left. - 5i \left(\frac{k_1/L}{\alpha_1^2}\right) \sum_{n=1}^{\infty} \frac{[J_1(k_1 Ln) + iY_1(k_1 Ln)]}{n} \cos(k_1 Ln \cos \theta_{\text{in}}) \right] / 5D_1 \alpha_1^3 \\ & + \text{higher-order terms,} \end{aligned} \quad (7.3)$$

in which

$$\bar{\alpha} = \alpha_1(1 + k_1^2/10\alpha_1^2). \quad (7.4)$$

The leading-order terms are identical to those of equation (5.21). The next-order term is an $O((k_1/\alpha_1)^2 \ln(\alpha_1/k_1))$ correction. However, this term is real and

contributes only to the higher-order correction to the resonant frequency. The next term is an $O((k_1/L)/\alpha_1^2)$ correction and its imaginary part determines the amplitude of the resonance. Thus,

$$\text{Im}(S_2) = \frac{k_1}{D_1 \alpha_1^5} \left[\frac{k_1 L}{4} + \sum_{n=1}^{\infty} \frac{J_1(k_1 L n)}{n} \cos(k_1 L n \cos \theta_{\text{in}}) \right] + \text{higher-order terms.} \quad (7.5)$$

The infinite summation remaining in equation (7.5) is evaluated in Appendix B, allowing the imaginary part of S_2 to be written finally as

$$\text{Im}(S_2) = \frac{1}{LD_1 \alpha_1^5} \sum_{l=-r}^m \sqrt{[(k_1 L)^2 - (k_1 L |\cos \theta_{\text{in}}| + 2l\pi)^2]} + \text{higher-order terms,} \quad (7.6)$$

where

$$2m\pi \leq k_1 L(1 - |\cos \theta_{\text{in}}|) < 2(m+1)\pi \quad (7.7)$$

and

$$2r\pi \leq k_1 L(1 + |\cos \theta_{\text{in}}|) < 2(r+1)\pi. \quad (7.8)$$

The reaction force amplitudes obtained from (3.11) for light rigid connectors and no fluid coupling are

$$F_1 = -F_2 = \frac{2L}{S_{11}(-k_1 \cos \theta_{\text{in}})(S_1 + S_2)} \quad (7.9)$$

and the plane wave propagating in the direction θ_n defined by (4.7) is obtained from equation (4.6) as

$$\frac{-2i\rho_1 \omega^2 \exp(ik_1 x \cos \theta_n) \exp(ik_1 z \sin \theta_n)}{S_{11}(-k_1 \cos \theta_{\text{in}}) S_{11}(k_1 \cos \theta_n) k_1 (S_1 + S_2) \sin \theta_n}. \quad (7.10)$$

At the resonant frequency where $S_1 + S_2 \approx 0$ this plane wave can be written as

$$\frac{-2L(\rho_1 \omega^2)^2 \exp(ik_1 x \cos \theta_n) \exp(ik_1 z \sin \theta_n)}{S_{11}(-k_1 \cos \theta_{\text{in}}) S_{11}(k_1 \cos \theta_n) k_1 \sin \theta_n \sum_{l=-r}^m \sqrt{[(k_1 L)^2 - (k_1 L |\cos \theta_{\text{in}}| + 2l\pi)^2]}} \quad (7.11)$$

where r and m are defined by (7.7) and (7.8), and which can be further approximated in the heavy fluid loading limit as

$$\frac{-2Lk_1 \sin \theta_{\text{in}} \exp(ik_1 x \cos \theta_n) \exp(ik_1 z \sin \theta_n)}{\sum_{l=-r}^m \sqrt{[(k_1 L)^2 - (k_1 L |\cos \theta_{\text{in}}| + 2l\pi)^2]}}. \quad (7.12)$$

Thus at combinations of frequency and incident angle that produce a resonance it is clear from this expression that to leading order the amplitude of the wave, which is $O(1)$, is independent of the direction of propagation, θ_n . Away from values of frequency and incident angle, which combine to produce a resonance, the leading-order term of $S_1 + S_2$ is S_1 as given by equations (5.11) and is therefore $O(1)$ with respect to ρ_1 . Thus away from the resonance the amplitude of the wave

propagating in the direction θ_n is $O(1/\rho_1)$. Therefore in the heavy fluid loading limit the acoustic pressure in the compressible fluid increases at resonance by a factor that is $O(\rho_1)$.

8. CONCLUDING REMARKS

The response of a system of two infinite parallel elastic plates with periodically spaced mechanical connectors and surrounded by compressible fluids, to an incident acoustic plane wave has been investigated by using Fourier transforms. The solutions can be decomposed as a plane wave scattered as if plate 1 were rigid together with a plane wave scattered in the same direction because of the vibration of the system of elastic plates and compressible fluids, which would occur in the absence of the mechanical connectors and terms that can be attributed directly to the presence of the mechanical connectors. The scattered pressure, which is caused by the connectors, consists of plane waves that propagate in a finite number of discrete directions, which depend on both the frequency of the incident wave and the angle of incidence. Formal expressions have been obtained for the wave amplitudes in terms of the physical parameters of the problem. These expressions are complicated, involving infinite sums of infinite integrals, but they could, however, be evaluated numerically to obtain solutions for particular parameter values. Herein both the integrals and the associated infinite sums have been evaluated approximately in the asymptotic limits of heavy fluid loading on plate 1, no fluid loading on plate 2 and either no fluid coupling between the plates or with heavy fluid between the plates and the additional limits of either large or small plate separation. A formal expression has also been obtained for the conditions necessary for a resonance (or free-wave propagation) in the system, and investigated further for light, rigid connectors in the asymptotic limits mentioned earlier. With no fluid coupling it was found that there are some frequency bands for which resonance never occurs. At frequencies where resonance is possible it occurs only when the incident angle is one of a finite number of values associated with that frequency. Formal expressions, to leading order, have been obtained for the amplitude of the scattered waves at resonance. These are $O(1)$, whereas the scattered wave amplitudes away from resonance are $O(1/\rho_1)$.

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APPENDIX A. HIGHER-ORDER ASYMPTOTIC EXPANSION OF $g(x)$ WITH HEAVY FLUID LOADING ON PLATE 1 AND NO FLUID COUPLING

(a) $x = 0$

When $x = 0$ the integrand of equation (5.14) is an even function of s , and $g(0)$ can be written as

$$g(0) = \frac{1}{\pi} \int_0^\infty \frac{ds}{D_1 s^4 - \omega^2 M_1 - i \rho_1 \omega^2 / \gamma_1(s)}. \quad (\text{A } 1)$$

The binomial theorem can be used to expand γ_1 , when $s > k_1$, as

$$\gamma_1(s) = is[1 - k_1^2/2s^2 + \dots], \quad s > k_1, \quad (\text{A } 2)$$

the first two terms of which can be used to approximate the integrand as

$$1/S_{11}(s) \approx s^3/(D_1 s^7 - \rho_1 \omega^2 s^2 - \rho_1 \omega^2 k_1^2/2), \quad s > k_1. \quad (\text{A } 3)$$

Equally, when $s < \alpha_1$ the binomial theorem can be applied to the integrand, allowing it to be expressed as

$$\frac{1}{S_{11}(s)} = \frac{i\gamma_1}{D_1 \alpha_1^5} \left\{ 1 + \left[\frac{(s^4 - \omega^2 M_1/D_1) \gamma_1}{i\alpha_1^5} \right] + \left[\frac{(s^4 - \omega^2 M_1/D_1) \gamma_1}{i\alpha_1^5} \right]^2 + \dots \right\}, \quad s < \alpha_1. \quad (\text{A } 4)$$

In the heavy fluid loading limit the regions in which these equations are valid overlap. By choosing S in this region as

$$S = k_1^{\frac{1}{2}} \alpha_1^{\frac{1}{2}}, \quad (\text{A } 5)$$

so that

$$k_1 \ll S \ll \alpha_1, \quad (\text{A } 6)$$

the range of integration for (A 1) can be split into regions where the integrand is well approximated by equation (A 3) or (A 4), i.e.

$$g(0) \approx \frac{1}{\pi} \int_0^S \frac{i\gamma_1}{D_1 \alpha_1^5} \left\{ 1 + \left[\frac{(s^4 - \omega^2 M_1/D_1) \gamma_1}{i\alpha_1^5} \right] + \left[\frac{(s^4 - \omega^2 M_1/D_1) \gamma_1}{i\alpha_1^5} \right]^2 + \dots \right\} ds \\ + \frac{1}{\pi} \int_S^\infty \frac{s^3 ds}{D_1 s^7 - \rho_1 \omega^2 s^2 - \frac{1}{2} \rho_1 \omega^2 k_1^2}. \quad (\text{A } 7)$$

The first integral can be evaluated term by term. For any $S \approx O(\alpha_1^q)$ where $q < 1$, which is satisfied by the choice (A 5), the first term of this integral includes $O(1/\alpha_1^5)$ components, whereas the integrals of the remaining terms are higher order and hence neglected here. Thus,

$$g(0) \approx (-k_1^2/2\pi D_1 \alpha_1^5) [(\alpha_1/k_1) - \frac{1}{2} \log(4\alpha_1/k_1) - \frac{1}{2} i\pi] \\ + \text{higher-order terms} + \frac{1}{\pi} \int_S^\infty \frac{s^3 ds}{D_1 s^7 - \rho_1 \omega^2 s^2 - \frac{1}{2} \rho_1 \omega^2 k_1^2}. \quad (\text{A } 8)$$

The integrand of the remaining integral is analytic except for seven simple poles. Five of these poles are located near to the corresponding poles found using only the leading-order approximation of γ_1 . These are located at

$$s \approx s_n, \quad n = 0, 1, 2, 3, 4, \quad (\text{A } 9)$$

where values of s_n are found to be

$$s_n = \alpha_1 \exp(\frac{2}{5}in\pi) (1 + k_1^2 \exp(-\frac{4}{5}in\pi)/10\alpha_1^2), \quad n = 0, 1, 2, 3, 4, \quad (\text{A } 10)$$

and the residue of the pole at s_n is A_n , where

$$A_n = (1 - k_1^2 \exp(-\frac{4}{5}in\pi)/2\alpha_1^2) \exp(\frac{4}{5}in\pi)/5D_1 \alpha_1^3, \quad n = 0, 1, 2, 3, 4. \quad (\text{A } 11)$$

From a consideration of the product of the pole locations, $\prod_{n=0}^6 s_n$, it is clear that the locations of the remaining two poles are 0(1) and thus they occur near to $s \approx \pm \frac{1}{\sqrt{2}} i k_1$. They are in fact located at

$$s \approx s_n, \quad n = 5, 6, \quad (\text{A } 12)$$

where

$$s_n = k_1 (\frac{1}{\sqrt{2}} i (-1)^n - \frac{1}{16} (k_1/\alpha_1)^5), \quad n = 5, 6, \quad (\text{A } 13)$$

and the residue at each of these poles is

$$A_n = k_1^2 / 4 D_1 \alpha_1^5, \quad n = 5, 6. \quad (\text{A } 14)$$

The integral can therefore be written in terms of the poles and residues as

$$\frac{1}{\pi} \int_S^\infty \frac{s^3 ds}{D_1 s^7 - \rho_1 \omega^2 s^2 - \frac{1}{2} \rho_1 \omega^2 k_1^2} = \frac{1}{\pi} \lim_{Q \rightarrow \infty} \int_S^Q \sum_{n=0}^6 \frac{A_n}{(s - s_n)} ds, \quad (\text{A } 15)$$

which can be evaluated as

$$\begin{aligned} \frac{1}{\pi} \int_S^\infty \frac{s^3 ds}{D_1 s^7 - \rho_1 \omega^2 s^2 - \frac{1}{2} \rho_1 \omega^2 k_1^2} &= \frac{i - \cot(\frac{2}{5}\pi)}{5 D_1 \alpha_1^3} \\ &+ (k_1^2 / 2 \pi D_1 \alpha_1^5) [(\alpha_1/k_1) + \frac{1}{2} \ln(\alpha_1/k_1) - \frac{1}{5} - \frac{1}{5} i \pi] \\ &+ \text{higher-order terms.} \end{aligned} \quad (\text{A } 16)$$

Thus

$$\begin{aligned} g(0) &= [i - \cot(\frac{2}{5}\pi) + (k_1/\alpha_1)^2 \{5 \ln(2\alpha_1/k_1) - 1 + \frac{3}{2} i \pi\} / 2\pi] / 5 D_1 \alpha_1^3 \\ &+ \text{higher-order terms.} \end{aligned} \quad (\text{A } 17)$$

(b) $x \neq 0$

When $x \neq 0$ the integration contour of equation (5.14) can be closed by a large semicircle in the upper half-plane when $x > 0$ or by a large semicircle in the lower half-plane when $x < 0$, allowing the integral to be expressed as the sum of residues and a branch cut integral. For $x \neq 0$ the residues of the complex poles are exponentially small in the heavy fluid loading limit and they can be neglected here. The location of the poles on the real axis that are crossed when the contour is deformed can be obtained from equation (A 10) as

$$s \approx \alpha_1 (1 + k_1^2 / 10 \alpha_1^2) \operatorname{sgn}(x) \quad (\text{A } 18)$$

and $g(x)$ can be written as

$$\begin{aligned} g(x) &\sim (1 - \frac{1}{2} (k_1/\alpha_1)^2) i \exp(i k_1 x \cos \theta_{\text{in}}) \exp(i \alpha_1 (1 + k_1^2 / 10 \alpha_1^2) |x|) / 5 D_1 \alpha_1^3 \\ &- \frac{i \rho_1 \omega^2 \exp(i k_1 x \cos \theta_{\text{in}})}{\pi} \int_{k_1}^\infty \frac{(e^{i s |x|} / \gamma_1) ds}{(D_1 s^4 - \omega^2 M_1 - i \rho_1 \omega^2 / \gamma_1) (D_1 s^4 - \omega^2 M_1 + i \rho_1 \omega^2 / \gamma_1)}, \end{aligned} \quad (\text{A } 19)$$

where the integral is to be evaluated on the left-hand side of the branch cut. This branch cut integral can be evaluated asymptotically by first dividing the

integration range at iS where S is defined by equation (A 5). It is straightforward to show that the integral from iS to $i\infty$ is exponentially small in the heavy fluid loading limit and may be neglected here. In the range from k_1 to iS the branch cut integrand can be expressed as

$$\int_{k_1}^{i\infty} \frac{(e^{is|x|}/\gamma_1) ds}{(D_1 s^4 - \omega^2 M_1)^2 + (\rho_1 \omega^2/\gamma_1)^2} \approx \frac{1}{(\rho_1 \omega^2)^2} \int_{k_1}^{iS} \gamma_1 e^{is|x|} \times \left\{ 1 - \left[\frac{(s^4 - \omega^2 M_1/D_1) \gamma_1}{\alpha_1^5} \right]^2 + \dots \right\} ds. \quad (\text{A } 20)$$

This integral can be evaluated term by term. The first term of the integral includes $O(1/\rho_1)$ contributions to $g(x)$, whereas the remaining terms are higher order and are neglected here. To evaluate this integral it is convenient to increase the upper limit of the integration range to $i\infty$. This adds a term which is exponentially small in the heavy fluid loading limit, which may be neglected. Thus

$$-i\rho_1 \omega^2 \exp(ik_1 x \cos \theta_{\text{in}})/\pi \int_{k_1}^{i\infty} \frac{(e^{is|x|}/\gamma_1) ds}{(D_1 s^4 - \omega^2 M_1)^2 + (\rho_1 \omega^2/\gamma_1)^2} \sim \frac{-i \exp(ik_1 x \cos \theta_{\text{in}})}{\pi \rho_1 \omega^2} \int_{k_1}^{i\infty} \gamma_1 e^{is|x|} ds \quad (\text{A } 21)$$

Integration by parts of this integral allows it to be written as

$$\frac{-i \exp(ik_1 x \cos \theta_{\text{in}})}{\pi \rho_1 \omega^2} \int_{k_1}^{i\infty} \gamma_1 e^{is|x|} ds = \frac{-\exp(ik_1 x \cos \theta_{\text{in}})}{\pi \rho_1 \omega^2 |x|} \int_{k_1}^{i\infty} \frac{s}{\gamma_1} e^{is|x|} ds, \quad (\text{A } 22)$$

which can be expressed in terms of the values on the opposite side of the branch cut, and thus deformed onto the positive real axis. By making the substitution for values of s on the real axis

$$s = k_1 y \quad (\text{A } 23)$$

this becomes

$$\frac{-i \exp(ik_1 x \cos \theta_{\text{in}})}{\pi \rho_1 \omega^2} \int_{k_1}^{i\infty} \gamma_1 e^{is|x|} ds = \frac{-ik_1 \exp(ik_1 x \cos \theta_{\text{in}})}{\pi \rho_1 \omega^2 |x|} \times \int_1^\infty \frac{y(\cos(k_1 y|x|) + i \sin(k_1 y|x|)) dy}{\sqrt{(y^2 - 1)}}. \quad (\text{A } 24)$$

Expressions for similar integrals are available in Gradshteyn & Ryzhik (1980, p. 419):

$$\int_1^\infty \frac{\cos(ax)}{\sqrt{(x^2 - 1)}} dx = -\frac{1}{2}\pi Y_0(a) \quad (\text{A } 25)$$

and

$$\int_1^\infty \frac{\sin(ax)}{\sqrt{(x^2 - 1)}} dx = \frac{1}{2}\pi J_0(a), \quad (\text{A } 26)$$

where J_0 and Y_0 are zero-order Bessel functions of the first and second kinds respectively. These equations can be differentiated with respect to a to obtain

expressions for the integrals of equation (A 24). Thus the leading-order terms of $g(x)$ are

$$\begin{aligned} g(x) = & i \exp(i k_1 x \cos \theta_{\text{in}}) [(1 - \frac{1}{2}(k_1/\alpha_1)^2) \exp(i \alpha_1(1 + k_1^2/10\alpha_1^2)|x|) \\ & + (5k_1/2\alpha_1^2|x|)(J_1(k_1|x|) + iY_1(k_1|x|))] / 5D_1 \alpha_1^3 \\ & + \text{higher-order terms, } x \neq 0. \end{aligned} \quad (\text{A } 27)$$

APPENDIX B. THE SUM OF A SERIES OF BESSEL FUNCTIONS

The infinite summation required in equation (7.5) can be evaluated by a further application of Poisson's summation theorem, by first writing it as

$$\sum_{n=1}^{\infty} \frac{J_1(k_1 L n)}{n} \cos(k_1 L n \cos \theta_{\text{in}}) = -\frac{1}{4}Q + \frac{1}{2} \sum_{n=-\infty}^{\infty} \frac{J_1(Qn)}{n} e^{-iQnt}, \quad (\text{B } 1)$$

where

$$Q = k_1 L \quad (\text{B } 2)$$

and

$$t = |\cos \theta_{\text{in}}|. \quad (\text{B } 3)$$

Poisson's summation theorem can now be applied with

$$g(x) = J_1(Qx) e^{-iQxt/2x}, \quad (\text{B } 4)$$

so that

$$G(s) = \int_{-\infty}^{\infty} \frac{J_1(Qx)}{2x} e^{-i(Qt+s)x} dx \quad (\text{B } 5)$$

or

$$G(s) = \int_0^{\infty} J_1(Qx) \cos((Qt+s)x) \frac{dx}{x}. \quad (\text{B } 6)$$

An expression for integrals of this form is provided by Gradshteyn & Ryzhik (1980, p. 743):

$$\int_0^{\infty} J_{\nu}(\alpha x) \cos(\beta x) \frac{dx}{x} = \begin{cases} \frac{1}{\nu} \cos\left(\nu \arcsin\left(\frac{\beta}{\alpha}\right)\right), & \beta \leq \alpha, \\ \frac{\alpha^{\nu} \cos(\frac{1}{2}\nu\pi)}{\nu(\beta + \sqrt{(\beta^2 - \alpha^2)})^{\nu}}, & \beta \geq \alpha. \end{cases} \quad (\text{B } 7)$$

This allows the evaluation of (B 6) as

$$G(s) = \begin{cases} (1/Q) \sqrt{[Q^2 - (Qt+s)^2]}, & |Qt+s| \leq Q, \\ 0, & |Qt+s| \geq Q. \end{cases} \quad (\text{B } 8)$$

Application of the Poisson summation theorem requires the evaluation of $G(2n\pi)$, which is non-zero only if

$$-Q(1+t) < 2n\pi < Q(1-t), \quad (\text{B } 9)$$

from which it may be deduced that

$$\frac{1}{2} \sum_{n=-\infty}^{\infty} \frac{J_1(Qn)}{n} e^{-iQnt} = \frac{1}{Q} \sum_{n=-r}^m \sqrt{[Q^2 - (Qt + 2n\pi)^2]}, \quad (\text{B } 10)$$

where

$$2m\pi \leq Q(1-t) < 2(m+1)\pi \quad (\text{B } 11)$$

and

$$2r\pi \leq Q(1+t) < 2(r+1)\pi. \quad (\text{B } 12)$$

Thus the summation required in equation (7.5) as

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{J_1(k_1 L n)}{n} \cos(k_1 L n \cos \theta_{\text{in}}) \\ = \frac{-k_1 L}{4} + \frac{1}{k_1 L} \sum_{l=-r}^m \sqrt{[(k_1 L)^2 - (k_1 L |\cos \theta_{\text{in}}| + 2l\pi)^2]} \end{aligned} \quad (\text{B } 13)$$

in which

$$2m\pi \leq k_1 L(1 - |\cos \theta_{\text{in}}|) < 2(m+1)\pi \quad (\text{B } 14)$$

and

$$2r\pi \leq k_1 L(1 + |\cos \theta_{\text{in}}|) < 2(r+1)\pi. \quad (\text{B } 15)$$

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